



Developing an Agentic AI System for Pathway and Strain Engineering

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SUMMARY

Metabolic engineering is transitioning toward data-intensive predictive design. However, the field remains hindered by fragmented computational toolchains and time-intensive manual orchestration. In recent years, Large Language Models (LLMs) have demonstrated substantial potential in automating complex scientific reasoning and facilitating natural language interaction with specialized knowledge bases. However, general-purpose LLMs often produce “biological hallucinations”—the generation of scientifically inaccurate biochemical data due to the absence of intrinsic physical and chemical constraints.

We developed an automated pipeline for agentic AI-assisted metabolic engineering that aims to streamline selected stages of the workflow by bridging natural language intent with rigorous computational toolchains. This system aims to improve the reliability of AI-driven scientific applications by integrating deterministic validation mechanisms into the reasoning process. The architecture utilizes the LangGraph framework and a React architecture for task decomposition and multi-tool orchestration. We further introduced a Search-Verify-Compute reasoning protocol that prioritizes structured entity verification and physical computation over probabilistic generation. The agent integrates a multisource ecosystem, including KEGG, UniProt, and the MEResource dataset for host evaluation, with computational engines for thermodynamic analysis and enzymatic assessment.

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1. INTRODUCTION

1.1 Background

1.1.1 AI for Science and the Paradigm Shift in Synthetic Biology

AI for Science and metabolic engineering have emerged as central drivers of the global bioeconomy, green manufacturing, and the sustainable development of biomedicine. With the rapid rise of the “AI for Science” paradigm, synthetic biology is undergoing a profound transformation—from traditional experience-driven, trial-and-error experimentation toward data-intensive, model-based predictive design.[1]

In this context, the computational reprogramming of microbial metabolic networks to produce high-value chemicals, advanced materials, and complex pharmaceuticals has demonstrated substantial industrial potential.[2] However, the inherent complexity of biological systems renders the precise regulation of intracellular metabolic fluxes exceedingly challenging. In the face of thousands of metabolites, biochemical reactions, and complex regulatory networks, the ability to rapidly, reliably, and cost-effectively identify optimal design strategies from an almost unbounded design space and to translate them into practical productivity remains a key strategic bottleneck limiting the large-scale application of synthetic biology.[2]

1.1.2 Current Status of Metabolic Engineering Design

Despite the emergence of numerous specialized computational tools for pathway discovery[3][4][5], thermodynamic evaluation[6][7], and enzyme mining[8][9], the design phase of cell factories remains highly fragmented from an industrial perspective.

Current metabolic engineering workflows rely heavily on manual orchestration by domain experts. Researchers are often required to switch frequently among incompatible software tools, heterogeneous databases, and complex bioinformatics

pipelines. This “manual assembly” paradigm presents several practical challenges. First, the lack of an integrated framework for tool execution and automated verification leads to high manual labor costs, as researchers need to devote substantial time to managing task transitions and cross-verifying the scientific accuracy of intermediate data. Second, scalability is limited: variability in individual expertise leads to poor reproducibility of design strategies, hindering the establishment of standardized and automated pipelines. Consequently, enhancing compatibility among computational tools and facilitating more integrated design workflows has become an important direction in both academic and industrial contexts.

1.1.3 The Potential of Large Language Models and Challenges

Generative AI technologies, represented by large language models (LLMs), offer a novel opportunity to address these interaction barriers. With strong capabilities in natural language understanding, logical reasoning, and code generation, LLMs have the potential to function as intelligent orchestration hubs for complex computational workflows, bridging human intent expressed in natural language with underlying computational tools.

However, the direct application of LLMs to rigorous scientific domains such as metabolic engineering faces a fundamental challenge—“biological hallucinations.” [10] Owing to the lack of intrinsic physical and biochemical constraints, general-purpose models may generate reactions that violate mass conservation or produce outputs that are not fully consistent with thermodynamic principles. In precision-sensitive contexts such as metabolic engineering, these limitations may affect the reliability and interpretability of model-generated suggestions, suggesting that purely probabilistic AI outputs still require careful validation before being used in experimental settings. Therefore, a key technical challenge in realizing the vision of an “AI scientist” lies in designing an agent architecture that combines the interactive flexibility of LLMs with the rigorous enforcement of scientific constraints.

1.2 Significance of This Study

To address the challenges associated with manual tool orchestration and the limited incorporation of scientific constraints in general-purpose AI models, this study presents a specialized agent-based system to support metabolic engineering workflows. The main contributions are as follows:

1. Facilitating tool orchestration through natural language interaction: We develop an agent architecture that enables researchers to interact with specialized computational tools using natural language. This approach helps streamline the use of bioinformatics pipelines and reduces the complexity of coordinating multiple tools.

2. Incorporating deterministic validation into the reasoning process: We introduce a “Search–Verify–Compute” (SVC) protocol that combines large language model reasoning with external data sources and computational engines. This framework allows intermediate results to be checked against biochemical constraints, improving the consistency and transparency of the system’s outputs.

3. Providing structured and evidence-informed design logic: The system supports task decomposition and generates structured responses grounded in tool-derived data and explicit verification steps. Instead of producing purely generative text, it offers a more traceable reasoning process that can assist researchers in evaluating design choices.

In summary, this work outlines a framework for integrating autonomous reasoning with established metabolic engineering tools, providing a foundation for more reliable AI-assisted computational design in synthetic biology.

1.3 Research Content and Objectives

To address the limitations of fragmented tools and the unreliability of general-purpose AI models, this study aims to develop an agentic framework designed to assist in computational metabolic engineering. The research is organized around four spe-

cific objectives, moving from the implementation of core reasoning logic to the systematic validation of the platform:

1. Implementation of a Task Orchestration and Decomposition Framework

This study aims to develop a system that can automatically break down complex metabolic engineering requests into manageable steps. Built on the LangGraph framework, the agent uses a ReAct(Reasoning and Acting) architecture to coordinate different tasks, such as metabolite verification, host selection, and pathway mapping through the MEResource dataset. The system can also perform physical validation steps such as thermodynamic analysis and catalytic element ranking. The agent dynamically decides which tools to use based on the task.

2. Implementation of a Constrained “Search-Verify-Compute” (SVC) Protocol

To mitigate biochemical hallucinations and meet the accuracy requirements of scientific design, this research proposes the “Search-Verify-Compute” (SVC) reasoning framework. This framework enforces a deterministic constraint protocol that prioritizes entity verification and physical computation over probabilistic text generation. In this paradigm, the LLM functions as an orchestrator that maps natural language user inputs to structured databases for input verification before invoking specialized computational engines. This approach ensures that reaction equations, thermodynamic data, and enzyme sequences are grounded in verified scientific data and rigorous derivation.

3. Systematic Integration of Heterogeneous Biological Toolchains

This objective focuses on building a unified microservice interface to enable interoperability between disparate biological resources. This includes the integration of foundational databases such as KEGG for metabolite entity alignment and UniProt for enzyme sequences and functional annotations. It further incorporates domain-specific datasets, specifically MEResource for Genome-Scale Metabolic Models (GEMs) based host evaluation, as well as physical engines including dGPPredictor, eEquilibrator, and EnzRank for thermodynamic and catalytic assessment. The goal is to pro-

vide the agent with a comprehensive and standardized toolkit for multi-dimensional metabolic analysis.

4. Quantitative Benchmarking and Performance Validation To quantitatively assess the performance of AI agents in complex scientific tasks, a specialized evaluation benchmark was developed. This benchmark, comprising 140 rigorous test cases, was developed to evaluate core functionalities, including atomic tool invocation, multi-step collaboration, systemic robustness, and out-of-domain intent filtering. System performance is quantified through a “Judge LLM”, which provides an automated assessment of accuracy, reasoning logic, and response completeness. Furthermore, the performance of mainstream open-source models, such as Llama 3.1 and Qwen 2.5, is compared to identify and optimize the core reasoning engines for metabolic engineering applications.

2. LITERATURE SURVEY

2.1 Current Status of Related Research

2.1.1 Large Language Models and Agentic AI Systems

In recent years, artificial intelligence has been undergoing a transition from standalone large language models (LLMs) to agentic AI systems with capabilities for autonomous planning and execution. As highlighted in prior research, future biomedical discovery may increasingly be driven by “AI scientists,” defined as systems capable of hypothesis-driven reasoning and iterative learning, while seamlessly integrating AI models with biomedical computational tools and experimental platforms.[1]At present, the application of agent technologies in scientific research is evolving from “assistants” toward “collaborators” [1]. According to the Levels of Autonomy framework, most existing systems remain at Level 1, where agents function as research assistants[1]. For instance, systems such as ChemCrow[11] and AutoBa[12] can execute predefined tasks—such as chemical synthesis planning or multi-omics analysis—through chain-of-thought reasoning and tool invocation, but their capabilities are largely limited to narrowly scoped problems. Recent work is gradually moving toward Level 2, where agents begin to demonstrate collaborative characteristics, including the ability to autonomously plan experimental protocols, invoke multiple external tools, and iteratively refine hypotheses based on feedback, as exemplified by systems such as Coscientist[13]. In parallel, cognitive architectures have evolved from simple ReAct (reasoning–acting) paradigms to more sophisticated multi-agent systems, where different agents assume roles such as brainstorming, debating, and reasoning, thereby improving robustness and reducing decision risks. Nevertheless, despite early progress in domains such as chemical synthesis and medical question answering, there remains a lack of robust, domain-specific agent frameworks for metabolic engineering, particularly for strain design

tasks that require strict adherence to metabolic flux constraints and thermodynamic feasibility.

2.1.2 Computational Design Tools in Metabolic Engineering

In the field of metabolic engineering, computational tools are transitioning from isolated algorithms to more integrated platforms[14], but they largely remain at the stage of non-agent-based tool invocation. Significant progress has been made in data accumulation, with studies such as those by Kim et al. constructing large-scale GEMs repositories that provide comprehensive stoichiometric foundations for host evaluation and pathway design[3]. At the same time, specialized tools have matured across multiple subdomains. For example, pathway design tools such as MooSeeker[4] employ Pareto optimization to balance pathway length and theoretical yield; thermodynamic analysis tools such as dGPredictor[6] and eQuilibrator[7] enable the estimation of Gibbs free energy changes and assessment of reaction feasibility based on group contribution methods and curated thermodynamic data; and enzyme prediction tools such as EnzRank[8] and ReactZyme[9] leverage protein language models to match enzymes with candidate reactions. In addition, integrated platforms such as novoStoic 2.0[14] attempt to combine stoichiometric optimization, pathway construction, and thermodynamic validation within a unified framework. However, despite these advances, most existing tools function as static computational engines without capabilities for self-reflection or dynamic task orchestration. As a result, researchers must still manually transfer data across heterogeneous tools and interfaces, which limits the efficiency and scalability of iterative design processes.

2.1.3 Limitations of Existing Studies

Despite these advances, several limitations persist. On the one hand, general-purpose LLMs applied to biochemical tasks may produce outputs that require careful valida-

tion, due to the lack of explicit physicochemical constraints. On the other hand, most existing computational tools remain isolated “point solutions”. Even integrated platforms typically lack natural language interfaces and automated task orchestration, requiring researchers to manually coordinate workflows across multiple systems. These limitations highlight the need for end-to-end, interactive AI agents that integrate LLM capabilities with domain-specific tools, in order to streamline iterative GEMs analysis and metabolic pathway design.

2.2 Theoretical Foundations

2.2.1 Large Language Models and Agent Technology

Large Language Models (LLMs) serve as the foundational computational engines for complex reasoning and natural language processing tasks. Built upon the Transformer architecture, these models utilize self-attention mechanisms to capture intricate semantic features and logical relationships within extensive datasets. Functionally, LLMs operate through token prediction, where the probability of subsequent semantic units is calculated based on a provided context. This mechanism enables robust natural language understanding, cross-disciplinary knowledge synthesis, and automated code generation. However, as probabilistic engines, standalone LLMs are primarily designed for pattern recognition and text synthesis; they lack inherent grounding in physical laws or deterministic logic, which necessitates the use of structured architectures when they are applied to rigorous scientific domains. Such architectures, commonly classified as Augmented Language Models, function as deterministic frameworks that decouple linguistic reasoning from technical execution. By validating information against authoritative databases and delegating complex calculations to specialized engines, these systems ensure that model outputs are grounded in empirical facts and physical laws rather than probabilistic estimation [11][15].

The integration of AI Agents represents an evolution from passive text gener-

ation to goal-directed execution. An Agentic system utilizes the LLM as a central controller to perceive environments, maintain memory, and execute actions via external tools. The ReAct (Reasoning and Acting) paradigm provides a structured cognitive cycle for this process, decomposing complex objectives into a sequence of “Reasoning” (generating internal thought traces) and “Acting” (invoking external functions). This interleaving allows the system to process real-time feedback from tool outputs and adapt subsequent steps accordingly. Such a framework is particularly effective for coordinating multi-step workflows that require both the linguistic flexibility of an LLM and the procedural precision of specialized software.

To manage the inherent complexity of multi-stage workflows, state-machine-driven execution frameworks have emerged as a robust alternative to linear scripting. By representing agentic processes as directed graphs composed of nodes (representing discrete actions) and edges (defining logic-based transitions), these frameworks facilitate sophisticated state management. This architectural approach supports non-linear execution, including conditional branching and cyclical loops, which are essential for tasks requiring iterative refinement or error recovery. In such systems, the “state” serves as a persistent record of the execution lifecycle, ensuring that all contextual information, including intermediate data, candidate solutions, and computational results, is maintained and dynamically updated as the task progresses.

2.2.2 Metabolic Engineering and Genome-Scale Metabolic Models

Metabolic engineering focuses on the directed modification of microbial metabolic networks to convert low-cost feedstocks into high-value target chemicals. The central challenge lies in identifying optimal metabolic pathways and matching them with highly efficient catalytic enzyme components. Effective design must not only satisfy mass balance requirements but also maintain competitive advantages in terms of energy supply (ATP) and cofactor balance (NADH/NADPH).

The scientific basis for the system’s decision-making is rooted in Genome-Scale

Metabolic Models (GEMs), which serve as “digital twins” of biological systems. GEMs integrate all known biochemical reactions, transport processes, and Gene-Protein-Reaction (GPR) associations within a specific organism. This research utilizes Flux Balance Analysis (FBA) to provide a quantitative foundation. By applying linear programming under known biochemical constraints and environmental conditions, FBA calculates the Maximum Theoretical Yield of a product. By systematically simulating multiple GEMs, the system can quantify the production potential of different chassis organisms for specific products, providing the underlying logic for autonomous host selection.

3. METHODOLOGY AND RESULTS

3.1 Technical Infrastructure

To support high-throughput data retrieval and the concurrent invocation of computational tools alongside complex LLM inference, the system is deployed on a high-performance computing node, the S9 server provided by SynCTI. This infrastructure features multi-core Intel® Xeon® processors and high-capacity system memory, complemented by NVIDIA acceleration hardware with significant video memory (VRAM). This local deployment ensures data privacy and provides the low-latency response times necessary for industrial-grade applications, enabling the efficient execution of advanced open-source models such as Qwen 2.5.

The system is built on a modular software stack isolated within Apptainer (formerly Singularity) to resolve library dependency conflicts and ensure environment reproducibility. The Agent Core, developed using FastAPI and Uvicorn, interfaces with the Ollama inference engine, which hosts the Large Language Model locally, and manages logic orchestration through LangChain and LangGraph. This state-machine-driven architecture utilizes SQLite for persistent session management and maintains a synchronous execution model to ensure the logical consistency of multi-step reasoning tasks. Biological and chemical validation is performed by a specialized suite of integrated tools, while Pandas with PyArrow enable high-speed retrieval from the MEResource dataset. The computation layer further incorporates eQuilibrator and dGPredictor for thermodynamic analysis, alongside the EnzRank module powered by TensorFlow, Keras, and Scikit-learn for enzyme evaluation. This pipeline is accessible via a Streamlit interface, which effectively connects high-level natural language intent with deterministic scientific computation.

3.2 System Architecture and Core Design

This section describes the design and implementation of the autonomous agentic system developed for metabolic engineering tasks. The architecture is based on the principle of decoupling the logical reasoning core from specialized computational tools. By utilizing a modular framework, a directed graph-based orchestration engine, and structured execution protocols, the system enables the automated coordination of metabolic engineering workflows. This approach facilitates multi-step design processes, such as pathway retrieval and thermodynamic validation, while improving the consistency of generated outputs through deterministic verification against biochemical databases.

3.2.1 Overall System Architecture

The system architecture, as illustrated in Figure.1, is organized into four distinct layers designed to facilitate the seamless synchronization of human intent with heterogeneous biological computational resources: the presentation layer, the logic orchestration layer, the data layer, and the computation layer. At the top of the architecture, the Presentation Layer utilizes a Streamlit-based Web UI (as shown in Figure. 2). The layout features a sidebar for session management and knowledge base navigation, a central stage displaying agent responses, and preset "Quick Input Templates" for frequently used modules like dGPredictor and EnzRank. This interface serves as the primary entry point for user interaction, translating natural language engineering objectives—such as specific pathway design goals—into actionable inputs for the core system without requiring specialized programming knowledge.

The Logic Orchestration Layer functions as the central control unit (labeled as "Orchestration" in the figure). This layer is driven by the Qwen-2.5:7b model, hosted locally via the Ollama framework. As shown in the left modules of the diagram, the reasoning process is governed by a set of rigorous System Prompts

and Logical Constraints. These protocols define the operational boundaries of the Agent, mandating that the model must invoke specific tools for biochemical validation rather than relying on probabilistic generation. The Data Layer, positioned to the right of the orchestration hub, provides the essential information required for accurate decision-making. It integrates a Short-Term Memory module for maintaining execution context and a Static Knowledge Base comprising KEGG, UniProt, and the MEResource dataset[3]. By grounding the system in these structured resources, the architecture ensures that all entity alignment and data retrieval processes are based on deterministic scientific facts.

The Computation Layer consists of the specialized bioinformatics toolkits shown at the bottom right. These include thermodynamic evaluation engines such as dG-Predictor and eQuilibrator, the EnzRank tool for enzymatic assessment, and dedicated modules for Pathway and Yield Search. These tools perform the final rigorous derivations and physical calculations necessary for metabolic design.

This layered architecture ensures high extensibility, allowing for the rapid integration of new computational models or databases through standardized tool nodes. This modular approach maintains the integrity of the system while providing the flexibility for evolving requirements.

3.2.2 Agent Cognitive Architecture Based on LangGraph

The system’s logical progression is modeled as a state machine utilizing the LangGraph framework, as illustrated in the directed graph in Figure.3. In contrast to traditional linear workflows, this graph-based architecture enables the Agent to perform iterative reasoning, self-correction, and multi-step planning. The cognitive cycle follows a structured sequence: Context Loading, Intent Reasoning, Decision Routing, and Tool Execution.

The process initiates at the Context Loader, which retrieves 10 previous dialogue exchanges from an SQLite Checkpointer (thread-scoped memory) and injects

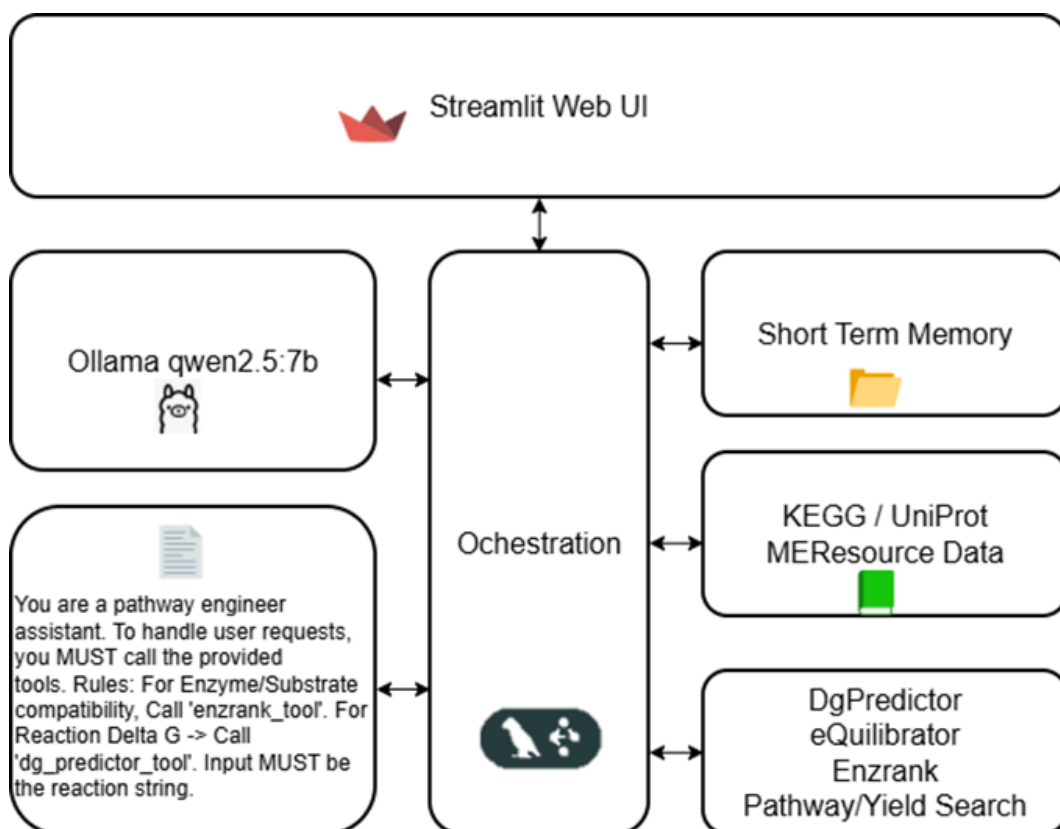


Figure 1: Overall system architecture of the autonomous metabolic engineering agent.

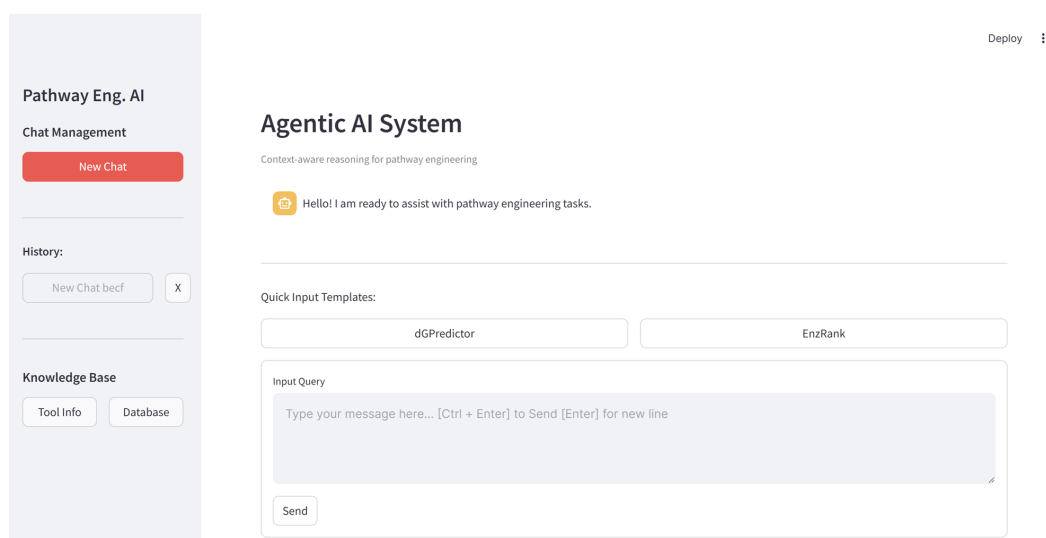


Figure 2: Overview of the integrated user interface for the Pathway Engineering AI Agent system.

scenario-specific system prompts. This ensures that the Agent maintains design consistency throughout multi-turn interactions—such as preserving specific host strain parameters or energy constraints across different queries. Following this, the Agent Node, powered by the Qwen-2.5 inference engine, processes the consolidated state to generate an execution strategy. It determines whether the current information is sufficient to resolve the user’s request or if external evidence is required.

Central to the architecture is the Decision Router, which implements a “should_continue” logic to manage the execution flow. If the Agent identifies a need for additional data, the router directs the process to the Tool Executor. This module manages the invocation of specialized Bio-informatics Toolkits, including dGPredictor, eQuilibrator, EnzRank, and various entity search utilities. To ensure robustness, the system utilizes both native tool-calling and a manual JSON fallback mechanism. This fallback uses regular expressions to locate and extract valid JSON blocks from the model’s raw text. This approach is particularly effective for handling non-standard outputs, such as when the model adds unnecessary conversational filler or makes minor formatting errors, ensuring that the agent can still execute tool calls without interruption. To ensure the traceability of the reasoning process, the Web UI provides an expandable “Tool Execution Logs” module, allowing users to inspect the underlying tool calls and their raw outputs (Figure. 5 in Appendices).

Once a tool execution is complete, the result is returned to the Agent Node as an Observation, allowing the model to refine its strategy based on empirical data. This iterative loop persists until the Decision Router determines that the task objectives have been met, at which point the system transitions to the final state to generate a structured Final Report. This state-machine-driven approach ensures that the design process is both transparent and logically rigorous.

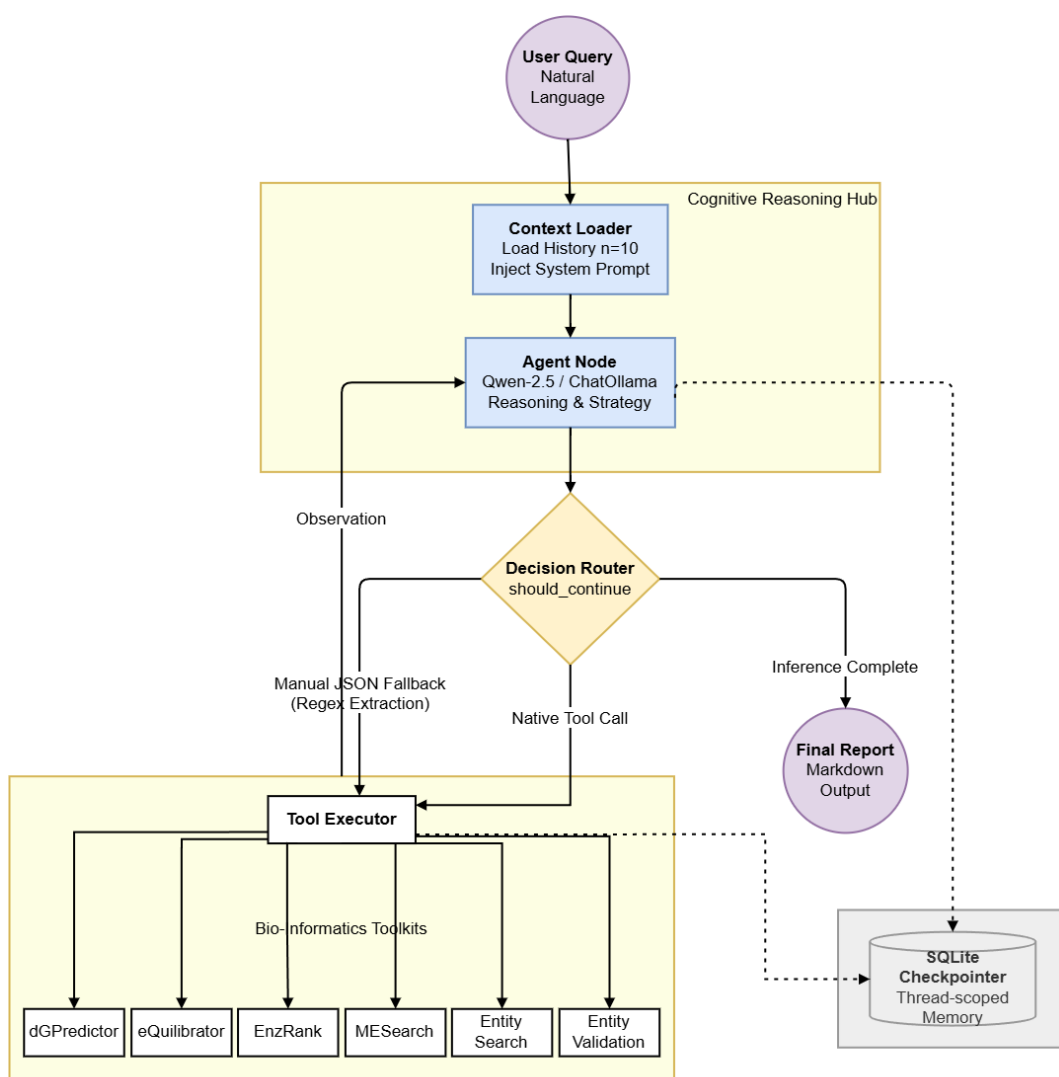


Figure 3: Flowchart of the agentic decision-making and execution process.

3.2.3 Deterministic Logic Protocols and Hallucination Mitigation

To meet the rigorous accuracy requirements of metabolic engineering, this research implements a “Search-Verify-Compute” (SVC) execution protocol, as illustrated in the sequence diagram in Figure. 4. This protocol functions as a deterministic constraint mechanism designed to eliminate the reliance on probabilistic generation for scientific data, ensuring that all design outputs are grounded in verified biochemical facts and physical laws. The SVC protocol is structured into three mandatory stages:

Protocol Step 1: Mandatory Decoding and Retrieval. Upon receiving a user request, the Agent is precluded from performing direct derivations based on internal prior knowledge. Instead, it must first invoke the `entity_validation_tool` to perform semantic grounding. As shown in the sequence (Figure. 4), natural language entities—such as “Fumarate”—are precisely mapped to standardized identifiers, such as KEGG Compound IDs (e.g., C00122). This stage ensures that the subsequent computational steps operate on unambiguous, high-fidelity biological entities rather than ambiguous natural language terms.

Protocol Step 2: Engine Selection and Execution. Once standardized identifiers are retrieved, the system identifies the optimal computational engine based on the task requirements. For thermodynamic validation, as exemplified in Figure. 4, the Agent transmits a balanced reaction string to the `equilibrator_tool`. By strictly aligning the input of physical models with structured data retrieved in the previous step, the system ensures that the resulting calculations are derived from deterministic scientific principles rather than LLM-based estimation.

Protocol Step 3: Result Synthesis and Reporting. In the final stage, the Agent aggregates the quantitative data returned by the computational engines—such as specific Gibbs free energy changes—and associates these values with the verified biochemical entities. The system then generates a structured output that summarizes the findings and design parameters. By presenting the calculated results alongside the standardized identifiers and formulas used, the protocol facilitates a data-driven

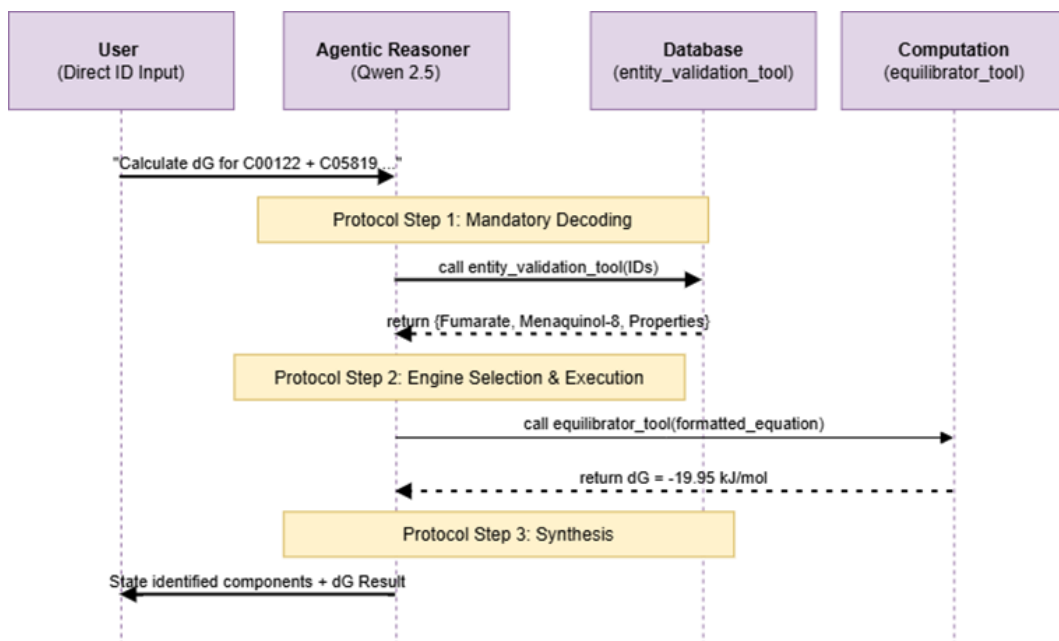


Figure 4: Operational sequence of the Search-Verify-Compute execution protocol

review process, allowing the user to trace the underlying computational evidence and data sources for each design component.

Furthermore, the system incorporates an Ask-back mechanism. If the initial user input contains critical omissions or logical inconsistencies, the Agent is programmed to interrupt the execution flow and solicit clarification from the user, to maintain a transparent and controllable environment, ensuring the integrity of the metabolic engineering design process.

3.2.4 Contextual Memory and Session Isolation Management

To support extended metabolic engineering design tasks, the system implements a persistent memory layer utilizing an SQLite-based checkpointing mechanism. This architecture ensures that the state of the design process is preserved across multiple interactions, facilitating long-term workflow continuity. Session integrity is maintained through a thread-scoped memory architecture. By isolating individual session streams via unique Thread IDs, the system ensures that state information remains encapsulated within its respective context. This prevents data cross-talk or state in-

interference in concurrent multi-user environments, ensuring that the parameters of one strain design do not affect another. The system’s ability to maintain design consistency across multi-turn queries is demonstrated in Figure. 6 in Appendices. The Agent successfully retains a long enzyme sequence in its short-term memory and pairs it with a substrate ID provided in a subsequent exchange to complete the calculation.

Furthermore, a sliding window technique is applied to manage the interaction history. To address the inherent context window limitations and potential attention decay of Large Language Models, the system prunes obsolete historical data and retains 10 most recent core dialogue exchanges. This strategy preserves the primary design objectives while effectively reducing the interference of redundant or irrelevant information.

3.3 Evaluation Setup

To systematically assess the Agent’s performance in complex metabolic engineering workflows, a benchmark of 140 test cases was constructed across five functional dimensions. Atomic Tool Invocation (65 cases) verifies the Agent’s ability to accurately utilize individual tools, such as dGPPredictor, EnzRank, and MESSearch, under explicit instructions. Complex Reasoning (10 cases) evaluates the orchestration of multi-step toolchains and the hybrid coordination of heterogeneous tools. This coordination involves managing sequential dependencies, where the output of one tool serves as the input for another, and conditional branching, where the agent determines the next tool to invoke based on intermediate results or specific task requirements. Additionally, Robustness and Boundary Defense (35 cases) tests defensive mechanisms against malformed biological identifiers or missing parameters. Out-of-Scope Intent Filtering (10 cases) assesses the ability to identify and reject non-bioengineering-related queries. Persona and System Awareness (20 cases) evaluates the Agent’s internal consistency and its recognition of operational boundaries.

The evaluation employs an “LLM-as-a-Judge” mechanism, utilizing the Llama 3.1 model to quantitatively score each system output on a scale of 1 to 5 across three primary metrics: Accuracy, which measures the correctness of tool selection and parameter extraction; Reasoning, which assesses the biological validity of the generated Chain-of-Thought (CoT) and logical rigor; and Completeness, which evaluates whether the final report incorporates all user-defined constraints. Each output is graded out of a maximum total of 15 points, and for this study, a Pass Rate is defined as a total score of $\geq 7/15$, provided that the output contains no critical scientific errors. This multi-dimensional assessment framework ensures the quantification of the Agent’s performance in real-world engineering scenarios.

3.4 Evaluation Results

3.4.1 Benchmarking LLM Performance

This study compared the performance of three prominent open-source models—Mistral-Nemo, Llama 3.1 (8B), and Qwen 2.5 (7B)—serving as the primary reasoning engine for the agentic system. Experimental results, summarized in Table 1, demonstrate that Qwen 2.5 (7B) exhibited superior performance across all evaluated dimensions.

Qwen 2.5 achieved a comprehensive average score of 12.37 ± 1.91 and a total pass rate of 96.4%. Further analysis indicates that the model’s high score in logical reasoning (4.29/5) is a decisive factor in its ability to navigate the complexities of the “Search-Verify-Compute” protocol. While Llama 3.1 and Mistral-Nemo demonstrated competitive capabilities, they frequently encountered issues such as parameter truncation during long enzyme sequence extraction and decision instability during multi-step tool routing. Based on these findings, Qwen 2.5 was selected as the default inference engine for the system.

Table 1: Performance comparison of different LLM reasoning engines.

Model Name	Accuracy (1–5)	Reasoning (1–5)	Completeness (1–5)	Avg. Total Score	Pass Rate (Score > 7)
Mistral-Nemo	3.23 ± 1.89	2.83 ± 1.70	3.43 ± 1.86	9.49 ± 4.87	69.30%
Llama 3.1 (8B)	3.79 ± 1.35	3.67 ± 1.15	3.93 ± 1.33	11.39 ± 3.06	88.60%
Qwen 2.5 (7B)	4.06 ± 0.96	4.29 ± 0.79	4.03 ± 1.17	12.37 ± 1.91	96.40%

3.4.2 Multi-Dimensional Capability Analysis

An analysis of Qwen 2.5 (7B) across 140 test cases reveals the system’s performance across three core dimensions: tool invocation, logical orchestration, and systemic robustness (Table 2). The detailed performance breakdown of Llama 3.1 and Mistral-Nemo across all 140 test cases is recorded in Table 3 and Table 4 in the Appendices, respectively. Comparative visualizations (Figs. 9–10 11 in the Appendices) also highlight Qwen 2.5’s superior stability across sub-tasks.

Qwen 2.5 achieved a 100% pass rate in atomic tool invocation tasks, specifically for *dGPredictor* ($n = 20$), *EnzRank* ($n = 20$), and *MESearch* ($n = 5$). In Database Retrieval ($n = 20$) tasks, despite complex nested JSON outputs, the system maintained an average score of 12.40/15. This reliability is largely due to the Manual JSON Fallback (Regex extraction) mechanism. In instances where the model generated non-standard formats, the system successfully extracted reaction equations via regular expressions, ensuring the consistent execution of underlying biochemical engines. This demonstrates the system’s stability in parsing high-fidelity parameters, such as KEGG IDs (e.g., C00122) and protein sequences.

The system maintained a 100% pass rate in the “Complex Reasoning and Multi-step Coordination” category, validating the “Search-Verify-Compute” (SVC) protocol. In multi-tool workflows, the agent demonstrated consistent logical chains: resolving a specific substrate to its standardized identifier, invoking `enzrank_tool` for enzymatic assessment, and proposing thermodynamic validation based on intermediate data. By requiring deterministic feedback from external tools at each stage, the system resolves nomenclature ambiguities that typically cause computational

errors in standalone LLMs, ensuring a scientifically rigorous reasoning process.

In 35 negative test cases involving invalid KEGG IDs or non-biological queries, the system exhibited robust defensive performance. Rather than generating probabilistic or fabricated responses, the Agent identified input errors and triggered an interactive clarification Ask-back strategy. Evaluation results show a 100% pass rate in the Neg_dG and Neg_DB categories. When encountering invalid identifiers or out-of-scope prompts, the Agent accurately recognized its operational boundaries and solicited correct information (e.g., requesting a valid KEGG ID). This explicit rejection mechanism effectively prevents the fabrication of biochemical data, providing a reliable logical defense for industrial applications.

Table 2: Detailed Performance Evaluation of Qwen 2.5:7b

Category	Sub-task	Acc.	Reas.	Compl.	Total	Pass	N
1. Atomic Invocation	dGPredictor	4.00 $\pm$.73	4.55 $\pm$.60	3.65 $\pm$.99	12.2 $\pm$.77	100%	20
	EnzRank	4.30 $\pm$.47	4.10 $\pm$ 1.0	3.95 $\pm$ 1.2	12.4 $\pm$ 1.4	100%	20
	DB Retrieval	3.95 $\pm$.94	4.15 $\pm$ 1.0	4.30 $\pm$ 1.1	12.4 $\pm$ 2.4	95%	20
	MESearch	4.20 $\pm$.45	3.60 $\pm$.89	4.60 $\pm$.55	12.4 $\pm$.55	100%	5
2. Complex Reasoning	Same-Tool	4.20 $\pm$.45	4.60 $\pm$.89	4.60 $\pm$.89	13.4 $\pm$ 1.3	100%	5
	Multi-Tool	4.40 $\pm$.55	4.20 $\pm$.84	4.20 $\pm$ 1.1	12.8 $\pm$ 1.1	100%	5
3. Robustness Defense	Neg_dG	4.00 $\pm$ 1.3	4.50 $\pm$.53	4.20 $\pm$ 1.3	12.7 $\pm$ 1.4	100%	10
	Neg_Enz	4.40 $\pm$.97	4.20 $\pm$.63	3.60 $\pm$ 1.3	12.2 $\pm$ 2.4	90%	10
	Neg_DB	4.20 $\pm$.92	4.50 $\pm$.53	4.20 $\pm$ 1.0	12.9 $\pm$ 1.7	100%	10
	Neg_ME	4.20 $\pm$.84	4.40 $\pm$.55	4.20 $\pm$.84	12.8 $\pm$.84	100%	5
4. Filtering	Out-of-Scope	3.20 $\pm$ 1.2	4.30 $\pm$.67	3.40 $\pm$ 1.6	10.9 $\pm$ 3.0	80%	10
5. Awareness	Greeting	3.50 $\pm$ 1.8	4.00 $\pm$.82	4.30 $\pm$ 1.6	11.8 $\pm$ 3.4	90%	10
	Capability Ask	4.50 $\pm$.53	4.50 $\pm$.53	4.10 $\pm$.99	13.1 $\pm$ 1.3	100%	10
Summary	Overall Avg	4.06 $\pm$.96	4.29 $\pm$.79	4.03 $\pm$ 1.1	12.4 $\pm$ 1.9	96.4%	140

3.4.3 Case Study: Succinate-Producing Cell Factory

This section demonstrates orchestration through the design and validation of a cell factory for succinate production. As illustrated in Figure. 8 in Appendices, when prompted with the query, “*I want to produce succinate. What is the best host?*”, the Agent invoked the *MESearch* module to query the pre-computed dataset. It identified *Pseudomonas putida iJN1463* as the optimal host and generated a D-glucose-based production pathway, utilizing FRD (fumarate reductase) and PFK (phospho-

fructokinase) to convert D-glucose to succinate via fumarate reduction.

In addition to providing the net equation, the Agent utilized its reasoning logic to identify specific metabolic constraints within the proposed designs. As shown in Figure. 8, it is noted that Pathway 2, a candidate route retrieved from the MERe-source dataset involving fumarate reductase (FRD) and an ADP-dependent kinase (PFK_adp), might exhibit lower energy efficiency compared to ATP-dependent alternatives. While the pathway's stoichiometric data was extracted from the database, the insight regarding its energetic limitations was generated by the Agent's reasoning engine. Based on this analysis, the system suggested thermodynamic verification as a necessary subsequent step to further evaluate the pathway's viability.

Upon a follow-up inquiry regarding reaction feasibility, the Agent focused on the specific conversion: fumarate + menaquinol-8 $\rightarrow$ menaquinone-8 + succinate. Initially, the system executed the mapping stage of the SVC protocol, precisely resolving the natural language terms into their standardized KEGG identifiers (e.g., C00122, C05819, C00828, and C00042) to eliminate nomenclature ambiguity (Figure. 7 in Appendices). Using these verified entities, the Agent then formulated a balanced reaction string to invoke the *dGPredictor* engine for rigorous thermodynamic evaluation. The calculation yielded a ΔG of -19.95 ± 0.16 kJ/mol, confirming that the reaction is thermodynamically favorable under standard physiological conditions. This process completed an evidence-driven design loop in seconds, demonstrating the system's utility in rapid metabolic engineering workflows.

3.4.4 Limitations

While the system demonstrated high performance in most scenarios, the evaluation identified specific limitations, particularly regarding the recognition of out-of-scope intent. In tests involving non-biological queries, the system achieved an 80% pass rate. In some instances, the model attempted to map unrelated terms to biological entities during database retrieval. These findings suggest that future iterations

should incorporate more rigorous persona alignment and static filtering logic. Such enhancements will ensure the system remains specialized and focused on metabolic engineering tasks, even when presented with high levels of interference or irrelevant input.

4. CONCLUSION

This study explored an integrated approach to streamlining metabolic engineering workflows by developing a prototype agentic system. By connecting Large Language Models with specialized bioinformatics tools, the system aims to facilitate the transition from natural language instructions to structured design reports, potentially reducing the need for manual data orchestration. The primary components and findings of this research are summarized below:

First, a reasoning architecture was implemented using the LangGraph framework to manage task decomposition. The inclusion of a sliding window memory mechanism helped maintain contextual consistency during multi-turn interactions.

Second, to address the issue of “biological hallucinations” of the general AI, this study introduced a “Search-Verify-Compute” execution workflow. By cross-referencing model-generated text with multiple databases and physical computation models, the system promotes greater consistency in the stoichiometric and thermodynamic parameters provided in the design suggestions.

Finally, the system’s performance was evaluated using a benchmark of 140 test cases. Utilizing Qwen 2.5 as the inference engine, the agent achieved a 96.4% pass rate under the defined criteria. In a succinate-producing cell factory case study, the system completed the specific design and validation steps within seconds, indicating its potential as a rapid intelligent assistant.

In conclusion, the experimental results suggest that such an integrated framework can help lower technical barriers in strain design while improving the traceability of the design process. Rather than replacing expert intervention, this research offers a practical workflow for deploying AI as a supportive tool to assist researchers within the broader Design-Build-Test-Learn cycle in synthetic biology.

5. RECOMMENDATION

While the current framework demonstrates the efficacy of LLM-driven metabolic design, its reliance on pre-defined datasets and APIs inherently constrains the accessible design space. To enhance the system’s applicability, it is recommended that the MEResource database be expanded to encompass a broader spectrum of non-model organisms, alternative carbon sources, and diverse bio-based products. For more complex design scenarios, such as *de novo* pathway construction, the integration of dynamic flux balance analysis (FBA) platforms, for instance, *novoStoic*—is proposed. This would enable real-time calculations of global stoichiometry and atomic conservation, moving beyond static data retrieval.

Furthermore, the introduction of multi-objective optimization algorithms is suggested to assist the Agent in balancing competing engineering goals, such as theoretical yield, metabolic burden, and intermediate toxicity. Such an approach could facilitate the generation of Pareto-optimal designs tailored for industrial-scale implementation. At the enzymatic level, the incorporation of generative protein language models is recommended to transition the system from evaluating known sequences toward predicting substrate specificity and identifying precise mutation sites for directed enzyme evolution.

To transition the system from a digital assistant toward a specialized “closed-loop AI scientist”, several architectural advancements are suggested. The integration of Retrieval-Augmented Generation (RAG) is recommended to enable the parsing of unstructured scientific literature from repositories such as PubMed and bioRxiv. This capability would allow the Agent to formulate hypotheses grounded in the most recent experimental findings, effectively mitigating the knowledge-lag issues inherent in static LLMs. Additionally, bridging short-term memory with Long-Term Memory (LTM) modules is proposed. By vectorizing and internalizing successful reasoning chains and expert feedback, the system could achieve a

form of continual learning, leveraging historical design experiences for novel tasks.

This architectural framework can also benefit from shifting from single-agent orchestration to a multi-agent system (MAS). It is recommended to introduce specialized agents—such as a “Generator” for proposing paths and a “Reviewer” for strict boundary validation—to improve the reliability of the design through internal discussion and self-correction. To support computationally intensive tasks such as dynamic FBA without compromising user interaction, an asynchronous orchestration architecture is suggested. Finally, the realization of embodied execution through connectivity with “Self-Driving Labs” is a critical future direction. Compiling structured design reports into low-level machine instructions for automated liquid handlers and cloud-based platforms would theoretically close the DBTL loop, bridging the gap between computational design and physical experimentation.

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APPENDICES

Table 3: Detailed Performance Evaluation of Llama 3.1

Category	Sub-task	Acc.	Reas.	Compl.	Total	Pass	N
1. Atomic Invocation	dGPredictor	5.00 ± 0.00	4.05 ± 0.22	4.95 ± 0.22	14.00 ± 0.00	100.0%	20
	EnzRank	3.35 ± 1.46	2.85 ± 1.53	3.50 ± 1.43	9.70 ± 3.71	70.0%	20
	Database Retrieval	4.35 ± 0.93	3.65 ± 1.27	4.05 ± 0.83	12.05 ± 1.82	95.0%	20
	MESearch	4.40 ± 0.55	3.20 ± 0.45	4.80 ± 0.45	12.40 ± 0.89	100.0%	5
2. Complex Reasoning	Same-Tool Batch	3.20 ± 1.48	3.80 ± 1.10	2.40 ± 1.82	9.40 ± 4.16	80.0%	5
	Multi-Tool Mixed	4.40 ± 1.34	3.80 ± 0.45	3.60 ± 0.89	11.80 ± 1.79	100.0%	5
3. Robustness Defense	Neg_dG	3.60 ± 1.07	4.00 ± 1.56	3.70 ± 1.64	11.30 ± 3.95	90.0%	10
	Neg_Enz	3.70 ± 1.25	4.10 ± 0.88	3.60 ± 1.26	11.40 ± 2.37	90.0%	10
	Neg_DB	3.90 ± 1.45	4.20 ± 0.63	4.60 ± 0.70	12.70 ± 1.57	100.0%	10
	Neg_ME	3.00 ± 1.41	3.20 ± 1.92	3.40 ± 1.52	9.60 ± 4.34	80.0%	5
4. Filtering	Out-of-Scope	2.20 ± 1.32	3.20 ± 1.23	3.10 ± 1.85	8.50 ± 3.63	70.0%	10
5. Awareness	Greeting	2.90 ± 1.29	3.90 ± 0.74	4.20 ± 1.14	11.00 ± 2.40	90.0%	10
	Capability Ask	3.90 ± 1.29	3.90 ± 0.88	3.70 ± 1.70	11.50 ± 3.14	90.0%	10
Summary	Overall Avg	3.79 ± 1.35	3.67 ± 1.15	3.93 ± 1.33	11.39 ± 3.06	88.6%	140

Table 4: Detailed Performance Evaluation of Mistral-Nemo

Category	Sub-task	Acc.	Reas.	Compl.	Total	Pass	N
1. Atomic Invocation	dGPredictor	4.75 ± 1.12	3.35 ± 1.23	4.50 ± 1.15	12.60 ± 3.17	95.0%	20
	EnzRank	3.05 ± 1.70	2.40 ± 1.73	3.30 ± 1.81	8.75 ± 4.51	65.0%	20
	Database Retrieval	2.55 ± 2.37	2.00 ± 1.89	2.55 ± 2.37	7.10 ± 6.49	50.0%	20
	MESearch	4.00 ± 0.00	3.20 ± 1.10	4.40 ± 0.89	11.60 ± 0.55	100.0%	5
2. Complex Reasoning	Same-Tool Batch	2.80 ± 2.17	2.40 ± 1.82	3.60 ± 2.19	8.80 ± 5.93	60.0%	5
	Multi-Tool Mixed	4.60 ± 0.55	3.60 ± 1.14	4.00 ± 1.00	12.20 ± 2.05	100.0%	5
3. Robustness Defense	Neg_dG	2.50 ± 1.96	3.00 ± 2.05	2.40 ± 2.07	7.90 ± 5.76	50.0%	10
	Neg_Enz	3.10 ± 1.97	2.50 ± 1.96	3.20 ± 1.99	8.80 ± 5.63	60.0%	10
	Neg_DB	3.40 ± 1.78	3.30 ± 1.57	3.50 ± 1.84	10.20 ± 4.54	80.0%	10
	Neg_ME	4.20 ± 1.30	3.20 ± 1.92	3.80 ± 0.84	11.20 ± 2.17	100.0%	5
4. Filtering	Out-of-Scope	1.70 ± 1.49	1.80 ± 1.75	2.30 ± 2.11	5.80 ± 3.82	30.0%	10
5. Awareness	Greeting	1.90 ± 1.66	3.10 ± 1.60	3.80 ± 1.40	8.80 ± 3.79	60.0%	10
	Capability Ask	4.10 ± 1.29	4.20 ± 0.63	4.20 ± 1.48	12.50 ± 2.51	90.0%	10
Summary	Overall Avg	3.23 ± 1.89	2.83 ± 1.70	3.43 ± 1.86	9.49 ± 4.87	69.3%	140


 Calculate dG for reaction: C00022 + C00004 + C00080 <=> C00186 + C00003


 View Tool Execution Logs (Internal)

```

?? Tool Called: dg_predictor_tool
Args: {'reaction_equation': 'C00022 + C00004 + C00080 <=> C00186 + C00003'}

? Result (dg_predictor_tool):
Success. Delta G = -27.55 kJ/mol (Std Dev: 0.10)...
```

The calculated standard Gibbs free energy change (ΔG°) for the reaction:

C00022 + C00004 + C00080 <=> C00186 + C00003

is approximately **-27.55 kJ/mol**, with a standard deviation of 0.10 kJ/mol.

A negative ΔG° indicates that the reaction is spontaneous under standard conditions, which means it will proceed in the forward direction without any external input of energy.


 Please calculate the EnzRank compatibility score for: Enzyme:
 MTKRVLVTGGAGFLGSHLCERLLSEGHEVICLDNFGSGRRKNIKEFHPSFKVNDNRDVRISESLPSVDRIYHLASRASPADFTQFPVNIALANTQGTRRLDQARA
 CDARMVFASTSEVYGDPKVHPQPETYTGNNIRGARGCYDESKRFGETLTVAQQRKYDVDARTVRFNTYGPMPDDGRVPTFVTQALRGDDLTIGDGEQTRS
 FCYVDDLIEGLISLMRVDNPEHNVYNIGKENERTIKELAYEVLGLTDESIVYEPLPEDDPGQRRPDITRAKTELDWEPKISLREGLEDITITYFDN Substrate:
 C00149


 View Tool Execution Logs (Internal)

```

?? Tool Called: enzrank_tool
Args: {'data': [{'enzyme': 'MTKRVLVTGGAGFLGSHLCERLLSEGHEVICLDNFGSGRRKNIKEFHPSFKVNDNRDVRISESLPSVDRIYHLASRASPADFTQFPVNIALANTQGTRRLDQARA CDARMVFASTSEVYGDPKVHPQPETYTGNNIRGARGCYDESKRFGETLTVAQQRKYDVDARTVRFNTYGPMPDDGRVPTFVTQALRGDDLTIGDGEQTRS FCYVDDLIEGLISLMRVDNPEHNVYNIGKENERTIKELAYEVLGLTDESIVYEPLPEDDPGQRRPDITRAKTELDWEPKISLREGLEDITITYFDN', 'substrate': 'C00149'}]}

? Result (enzrank_tool):
[{"protein_id": "AutoID", "compound_id": "C00149", "enzrank_score": 0.9314018487930298, "status": "success"}]
```

The EnzRank compatibility score for the given enzyme and substrate is 0.931. This indicates a high likelihood that the enzyme can catalyze the reaction with the specified substrate.

Figure 5: UI demonstration of the tool execution and traceability logs for dGPredictor and EnzRank tools.

 Please calculate the EnzRank compatibility score for: Enzyme:
 MTKRVLVTGGAGFLGSHLCERLLSEGHEVICLDNFGSGRRKNIKEFEDHPSFKVNDRDRISESLPSVDRIYHLASRASPADFTQFPVNIALANTQGTRRLDQARA
 CDARMVFASTSEVYGDPKVHPQPEITYTGNVNIRGARGCYDESKRFGETLVAYQRKYVDVARTVRIFNTYGPRMRPDDGRVVPFVTQALRGDDLTIGDGEQTRS
 FCYVDDLIEGLISLMRVDNPEHNVYNIGKENERTIKELAYEVLGLTDTESDIVYEPLPEDDPGQRRPDITRAKTELDWEPKISLREGLEDITITYFDN

 View Tool Execution Logs (Internal) 

I apologize for the error. It seems that I forgot to include the substrate information in the request. Could you please provide the substrate for which you want to calculate the EnzRank compatibility score?

 Use Substrate: C00149

 View Tool Execution Logs (Internal) 

```
?? Tool Called: enzrank_tool
Args: {'data': [{'enzyme': 'MTKRVLVTGGAGFLGSHLCERLLSEGHEVICLDNFGSGRRKNIKEFEDHPSFKVNDRDRISESLPSVDRIYHLASRASPADFTQFPVNIALANTQGTRRLDQARA
CDARMVFASTSEVYGDPKVHPQPEITYTGNVNIRGARGCYDESKRFGETLVAYQRKYVDVARTVRIFNTYGPRMRPDDGRVVPFVTQALRGDDLTIGDGEQTRS
FCYVDDLIEGLISLMRVDNPEHNVYNIGKENERTIKELAYEVLGLTDTESDIVYEPLPEDDPGQRRPDITRAKTELDWEPKISLREGLEDITITYFDN'}]}

? Result (enzrank_tool):
[{"protein_id": "AutoID", "compound_id": "C00149", "enzrank_score": 0.9314018487930298, "status": "success"}]
```

The EnzRank compatibility score for the given enzyme and substrate (C00149) is approximately 0.931.

Figure 6: UI demonstration of contextual memory and the Ask-back mechanism.

 Find the exact KEGG Compound IDs for fumarate(2-) + menaquinol-8 -> menaquinone-8 + succinate(2-)

 View Tool Execution Logs (Internal) 

Based on your search, here are the KEGG Compound IDs for the given compounds:

- Fumarate(2-) -> C00122
- Menaquinol-8 -> C05819
- Menaquinone-8 -> C00828
- Succinate(2-) -> C00042

So, the reaction with KEGG Compound IDs would be:

C00122 + C05819 → C00828 + C00042

Figure 7: UI demonstration of the entity mapping and KEGG ID alignment process.

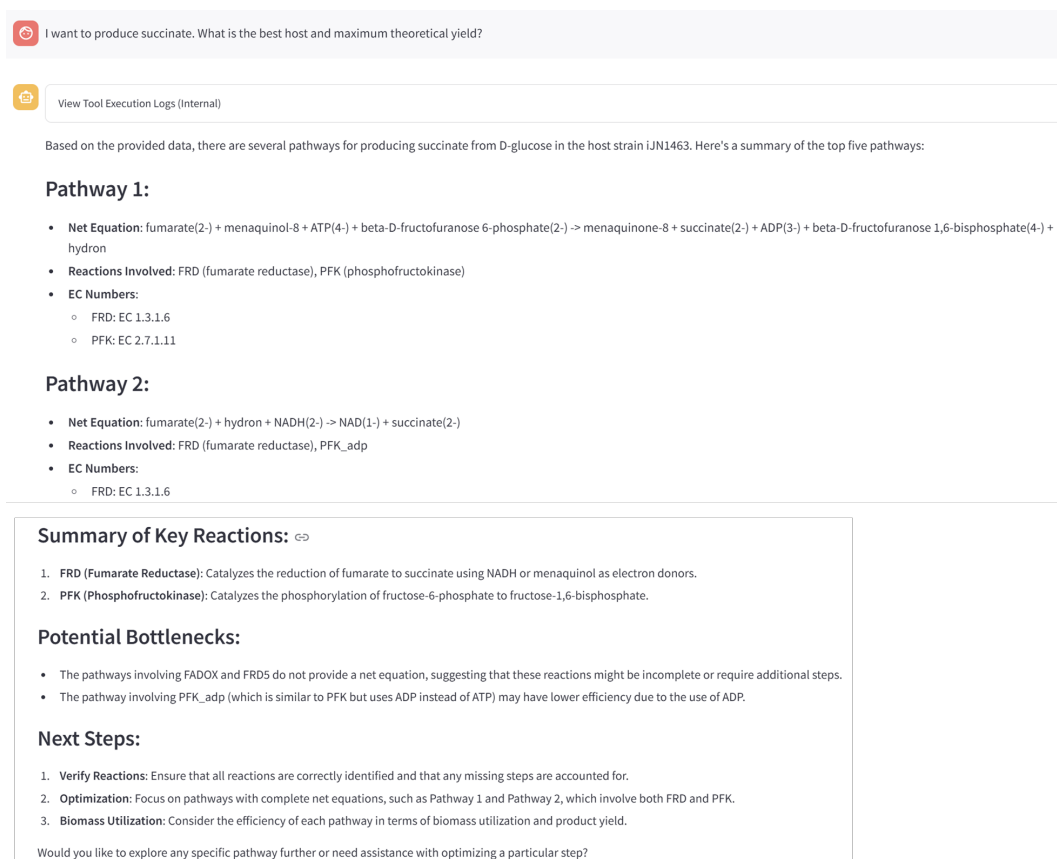


Figure 8: UI demonstration of the structured output generated by the Agent for the succinate-producing cell factory case study.

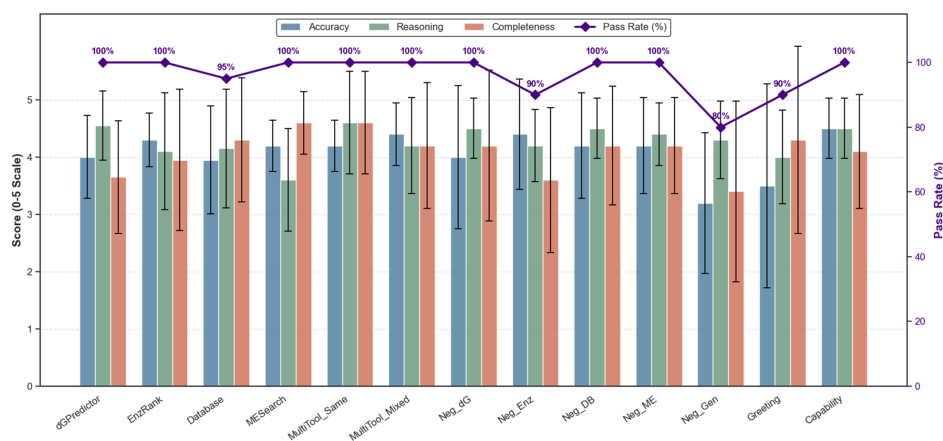


Figure 9: Multi-dimensional performance analysis for Qwen 2.5:7b.

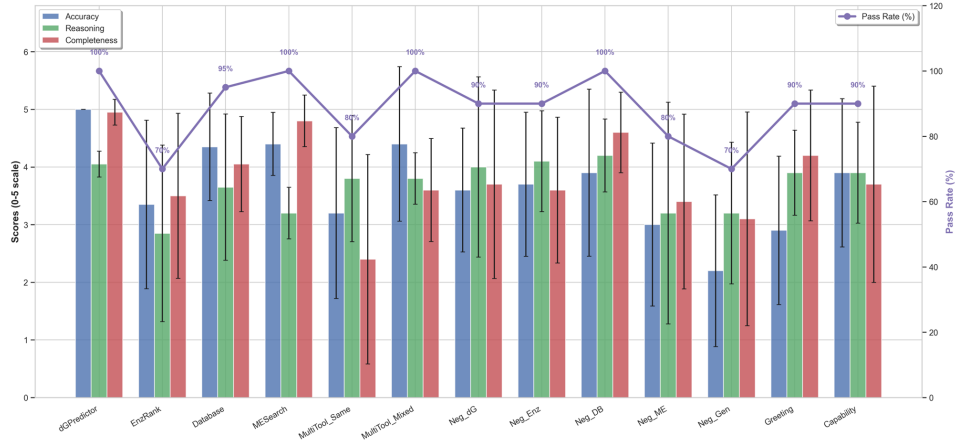


Figure 10: Agent performance benchmarking for Llama 3.1.



Figure 11: Agent performance benchmarking for Mistral-Nemo.